Hospital Management System: ER Diagram

TriageAssessments

- **PK triage_id**
- *FK admission_id*
- assessment_datetime
- triage_nurse
- blood_pressure
- heart_rate NUMERIC
- respiratory_rate NUMERIC
- body_temperature NUMERIC
- oxygen_saturation NUMERIC
- pain_level (0-10)
- reported_symptoms
- known_allergies
- current_medications
- priority_level (High / Moderate / Low)

ClinicalNotes

- **PK note_id**
- *FK admission_id*
- note_datetime
- author_physician
- note_type
- symptoms
- physical_examination
- clinical_findings
- suspected_condition
- planned_examinations
- treatment_plan
- note_content TEXT

UnitStays

- **PK unit_stay_id**
- *FK admission_id*
- unit_type (ICU, CCU, MICU, SICU, ...)
- ward
- room
- bed_number
- entry_datetime
- exit_datetime (>= entry_datetime)
- responsible_physician
- transfer_reason

RadiologyExaminations

- **PK radiology_exam_id**
- *FK admission_id*
- *FK order_id*
- exam_type
- exam_datetime
- body_area
- referring_physician
- radiology_technician
- exam_status
- image_link

reports on
(exam_id UNIQUE)

RadiologyReports

- **PK report_id**
- *FK radiology_exam_id (UNIQUE)*
- radiologist
- report_datetime
- findings
- interpretations TEXT
- clinical_conclusion

assessed in

documents

stays during

includes

requested by
composite FK (order_id, admission_id)

Patients

- **PK patient_id**
- first_name
- last_name
- date_of_birth
- gender
- address
- telephone_number
- health_insurance_number (UNIQUE)
- emergency_contact_name
- emergency_contact_phone
- life_status (Alive / Deceased)
- death_datetime TIMESTAMP (NULL if alive)

has

Admissions

- **PK admission_id**
- *FK patient_id*
- admission_datetime
- admission_type (CHECK)
- admission_source
- reason_for_admission
- insurance_info
- condition_on_admission
- admitting_professional
- discharge_datetime (>= admission_datetime)
- final_status (CHECK)

contains

DiagnosticOrders

- **PK order_id**
- *FK admission_id*
- requesting_physician
- requested_examination
- request_datetime
- clinical_reason
- priority (CHECK)
- order_status (CHECK)

produces

DiagnosticExaminations

- **PK exam_id**
- *FK order_id*
- exam_type
- exam_datetime (nullable)
- performed_by
- exam_result TEXT (nullable)
- exam_status (CHECK)

closes (admission_id UNIQUE)

diagnosed in
composite FK (patient_id, admission_id)

performed in

receives

Diagnosis

- **PK diagnosis_id**
- *FK patient_id*
- *FK admission_id*
- *FK icd9_code*
- diagnosis_datetime
- diagnosis_type (Primary / Secondary / Prelimi
- diagnosing_physician

coded as

ICD9Dictionary

- **PK icd9_code**
- official_title
- description
- disease_category

SurgicalProcedures

- **PK procedure_id**
- *FK admission_id*
- procedure_name
- procedure_datetime
- surgeon
- operating_room
- outcome_notes

DischargeRecords

- **PK discharge_id**
- *FK admission_id (UNIQUE)*
- discharge_datetime
- discharge_destination
- condition_at_discharge
- prescribed_medications
- follow_up_instructions
- follow_up_appointments
- responsible_physician
- discharge_summary